

# **Level 1 mini-project report**

## **STUDENT BURNOUT RISK PREDICTION USING MACHINE LEARNING**

### **1. Introduction**

Artificial Intelligence has become an important part of modern education. Students increasingly use AI tools for learning, assignments, research, and exam preparation. While these tools improve productivity and learning efficiency, excessive dependence on AI may also influence academic performance, study habits, and stress levels.

The objective of this project is to predict the burnout risk level of students using machine learning techniques. By analyzing various academic and behavioral factors, the model attempts to identify whether a student is at low, medium, or high risk of burnout.

### **2. Problem Statement**

Student burnout is a major concern in educational institutions. Factors such as study hours, anxiety levels, academic performance, AI dependency, and learning behavior can contribute to burnout.

The goal of this project is to develop a machine learning model capable of predicting the burnout risk level of students using the available dataset.

### **3. Dataset Description**

The dataset contains information about students, their academic performance, AI usage patterns, and psychological factors.

- Dataset Characteristics:
  - Total Records: 50,000
  - Total Features: 16
  - Target Variable: Burnout Risk Level
- Important Attributes:
  - Major Category
  - Year of Study
  - Pre-Semester GPA
  - Weekly GenAI Hours
  - Primary Use Case

- Prompt Engineering Skill
- Tool Diversity
- Paid Subscription
- Traditional Study Hours
- Perceived AI Dependency
- Institutional Policy
- Anxiety Level During Exams
- Post Semester GPA
- Skill Retention Score

The target variable classifies students into different burnout risk levels.

**Datasheet link:** <https://www.kaggle.com/datasets/laveshjadon/ai-impact-on-students>

#### 4. Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning.

The following preprocessing steps were carried out:

- Label Encoding:

Categorical attributes were converted into numerical values using Label Encoding. This transformation allows machine learning algorithms to process categorical information efficiently.

- Dataset Splitting:

The dataset was divided into training and testing sets.

- Training Data: 80%
- Testing Data: 20%

#### 5. Machine Learning Model

- Algorithm Used: Random Forest Classifier

Reason for Selection:

Random Forest is a powerful ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

- Advantages include:
  - Handles large datasets efficiently
  - Supports classification problems

- Provides stable predictions
- Reduces variance compared to a single decision tree

The model was trained using the processed dataset and then evaluated on the testing data.

## 6. Model Evaluation

The performance of the model was measured using Accuracy Score.

- Accuracy Obtained: 51.15%

Analysis:

The model successfully learned patterns from the dataset and achieved an accuracy of 51.15%.

The moderate accuracy indicates that burnout prediction is a complex problem influenced by numerous personal, academic, and environmental factors. Additional features and advanced machine learning techniques may further improve prediction performance.

## 7. Prediction Result

After training, the model was saved and later loaded for prediction.

A sample student record was provided to the model.

- Prediction Output :

Prediction = 2

Based on the label encoding used during training, this prediction corresponds to:

Medium Burnout Risk

This indicates that the student may experience a moderate level of academic stress and burnout.

## 8. Conclusion

This project successfully implemented a machine learning solution for predicting student burnout risk using educational and behavioral data.

The Random Forest Classifier was trained using 50,000 student records and achieved an accuracy of 51.15%.

The project demonstrated the complete machine learning workflow, including:

- Data loading
- Data preprocessing
- Label encoding
- Dataset splitting

- Model training
- Model evaluation
- Model saving
- Prediction

The developed model can assist educational institutions in identifying students who may require academic or psychological support.

## **9. Future Scope**

Several improvements can be made in future work:

- Collect additional student behavioral data.
- Explore advanced machine learning algorithms.
- Improve feature engineering techniques.
- Develop a real-time prediction system.
- Create a web-based application for burnout monitoring.

These enhancements may improve prediction accuracy and practical usability.

## **10. Learning Outcome**

Through this project, valuable knowledge was gained regarding machine learning workflows and predictive modeling.

The project provided hands-on experience in:

- Data preprocessing
- Feature transformation
- Classification algorithms
- Model evaluation
- Model deployment using Joblib

It also demonstrated how machine learning can be applied to solve real-world educational challenges and support student well-being.

```
L1_project.py > ...
6  import joblib
7
8  data = pd.read_csv("ai_student_impact_dataset.csv")
9
10 data = data.drop_duplicates()
11
12 label_encoder = LabelEncoder()
13
14 data["Major_Category"] = label_encoder.fit_transform(data["Major_Category"])
15 data["Year_of_Study"] = label_encoder.fit_transform(data["Year_of_Study"])
16 data["Primary_Use_Case"] = label_encoder.fit_transform(data["Primary_Use_Case"])
17 data["Prompt_Engineering_Skill"] = label_encoder.fit_transform(data["Prompt_Engineering_Skill"])
18 data["Institutional_Policy"] = label_encoder.fit_transform(data["Institutional_Policy"])
19 data["Burnout_Risk_Level"] = label_encoder.fit_transform(data["Burnout_Risk_Level"])
20
21 X = data.drop(["Student_ID", "Burnout_Risk_Level"], axis=1)
22 Y = data["Burnout_Risk_Level"]
23
24 X_train, X_test, Y_train, Y_test = train_test_split(X,Y,test_size=0.2,random_state=42)
25
26 model = RandomForestClassifier(n_estimators=100, random_state=42)
27 model.fit(X_train, Y_train)
28
29 Y_pred = model.predict(X_test)
30
31 accuracy = accuracy_score(Y_test, Y_pred)
32
33 print("Accuracy =", accuracy*100,"%")
34
35 joblib.dump(model, "student_burnout_model.pkl")
36 print("Model Saved Successfully")
37
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell + ▢ ▢ ... | 🔍 ×

PS C:\Users\ADITHYAN H\Downloads\LEVEL 1&gt; py l1\_project.py

Accuracy = 51.15 %

Model Saved Successfully

PS C:\Users\ADITHYAN H\Downloads\LEVEL 1&gt; ▢

```
predict.py > ...
1  import joblib
2
3  model = joblib.load("student_burnout_model.pkl")
4
5  import pandas as pd
6
7  sample = pd.DataFrame([[2,1,3.5,10,2,1,4,0,12,3,1,5,3.6,80]], columns=["Major_Category", "Year_of_Study", "Pre_Semester_GPA",
8  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
9  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
10 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
11 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
12 prediction = model.predict(sample)
13
14 print("Prediction =", prediction)
15
16 if prediction[0] == 0:
17     print("High Burnout Risk")
18 elif prediction[0] == 1:
19     print("Low Burnout Risk")
20 else:
21     print("Medium Burnout Risk")
```

```
PS C:\Users\ADITHYAN H\Downloads\LEVEL 1> py predict.py
Prediction = [2]
Medium Burnout Risk
PS C:\Users\ADITHYAN H\Downloads\LEVEL 1> 
```